Sarbajit Ghosh

CONTACT — RM511, CSE Department, IIT Kanpur

CAREER VISION — I look forward to utilizing my skills and ability in learning state-of-the-art cryptology, machine learning & data security, as well as solving challenging and innovative problems in this area. In this way, I like to deeply engage myself in advancing the subject to a better society.

SKILLS — Theoretical cryptology, Multi-party Computation, Blockchain & Cryptocurrency, Zero-knowledge proofs, Machine Learning, Bayesian Inference, Linear Algebra, Differential Privacy, etc.

ACADEMIC BACKGROUND

Ph.D (Pursuing) in Computer Science — 2021 - Present
Indian Institute of Technology, Kanpur.
CPI: 8.00

M.Tech in Cryptology and information security. — 2019 - 2021
Indian Statistical Institute, Kolkata.
Marks: 81.35%
First Division with Distinction.

GATE Qualified In Statistics. — 2019
Percentile: 98.01

GATE Qualified In Computer Science. — 2021
Percentile: 87.26

M.Sc (Post Graduation) in Statistics. — 2016 - 2018
West Bengal State University, Barasat.
Marks: 81.3%
Recipient of First-Class First Vice-Chancellor Gold Medal.

B.Sc(Graduation) in Mathematics. — 2012 - 2016
Rishi Bankim Chandra College, under West Bengal State University.
Marks: 54.63%

Higher Secondary Examination. — 2009 - 2011
Naihati Narendra Vidyaniketan, under WBCHSE
Marks: 72.6%

Secondary Examination. — 2009
Ramkrishna Vivekananda Misson, under WBBSE
Marks: 79.38%

PUBLICATIONS

1. A Generalized Unknown Malware Classification. Rani, N., Mishra, A., Kumar, R., **Ghosh, S.**, Shukla, S.K. and Bagade, P. International Conference on Security and Privacy in Communication Systems (SecureComm), 2022

TEACHING ACTIVITIES

1. **Teaching Assistant, CSE, IIT Kanpur**
   - CS689 Computational Linguistics for Indian Languages — Spring 2024, Fall 2022
   - CS798 Introduction to Probability for Computer Science — Fall 2023

- ESC112 Fundamentals of Computing - II Summer 2023
- CS315 Principles of Database Systems Spring 2023
- CS657 Information Retrieval Spring 2022
- CS685 Data Mining Fall 2021

2. **Guest Lecturer at the department of statistics, Chakdaha Day College.** August 2018 – July 2019
   - **Responsibilities:** Teaching, Organising internal & external examinations.
   - **Courses taught**: Descriptive statistics & probability theory, Statistical inference at UG level.

Internship

1. **Statistical Programmer at Imerit Technologies, Kolkata, India** June 2020 - August 2020
   - **Project Name:** An Analysis of Factors that Impact Data Annotation Productivity in Complex CV Projects.
   - **Project Domain: Statistical Analysis**, **Socio-economic Survey**.
   - **Description of Project:** To model to time per task, in CV projects of an employee, based on project characteristics and some external parameters such as training given by the company, socio-economic profile of that employee, etc as a part of a team.
   - **Technologies Used: R** packages **dplyr**, **ggpolt2**, **caret**, **Google sheets**, **Tableau**, NoSQL database **MongoDB**.

Academic Projects

1. **Internet voting using Blockchain.** Spring 2021
   Guide: Dr. Souradyuti Paul, IIT Bhilai.
   - **Project Domain: Secure Multiparty computation**, **Blockchain**, **Internet voting**.
   - **Description of Project:** Application of Ethereum blockchain, in designing a complete decentralized voting protocol together with related security proofs and zero-knowledge proofs.
   - **Technologies Used: Ganache**, **Truffle**,**MetaMask**, **Solidity**, **W3.JS**.

2. **A short survey on Differential Privacy.** Fall 2020
   Supervisor: Dr. Srimanta Bhattacharya, R. C. Bose Centre for Cryptology and Security.

3. **Secure Banking Website.** Spring 2020
   Supervisor: Dr. Nilanjan Datta, ISI Kolkata.
   Project Purpose: To create a banking website which is secure against **SQL injection**, **CSRF** and **Clickjack attack**.

4. **A note on How to Access Program Counter and Instruction Register by C programming.** Fall 2019
   Supervisor: Prof. Subhamoy Maitra, ISI Kolkata.

5. **Robust model-based adjustments to GREG estimation when some responses are missing.** Spring 2018
   Supervisor: Dr. Sumanta Adhya, WBSU.
   Project Purpose: To study the robustness property of GREG estimator in different cases when imputation was done by parametric and non-parametric regression and when data size is small.

Programming Skill

- **General:** C, Python, R, Excel.
- **Web Development:** HTML, PHP.
- **Database & Publishing:** SQL, LaTeX, Markdown.
- **Block Chain:** Solidity.
- **OS:** Linux.

Personal Details

- **Date Of Birth:** 01/June/1992.
- **Languages:** Bengali (Mother tongue), English, Hindi.
- **Gender:** Male.
- **Nationality:** Indian.
- **Hobbies:** Violin, Indian Classical Music, Regional History.